Week 10

[1]. Bruce, J., Dennis, M.D., Edwards, A., Parker-Holder, J., Shi, Y., Hughes, E., Lai, M., Mavalankar, A., Steigerwald, R., Apps, C., Aytar, Y., Bechtle, S., Behbahani, F.M., Chan, S., Heess, N.M., Gonzalez, L., Osindero, S., Ozair, S., Reed, S., Zhang, J., Zolna, K., Clune, J., Freitas, N.D., Singh, S., & Rocktaschel, T. (2024). Genie: Generative Interactive Environments. ArXiv, abs/2402.15391.

This is a new form of generative AI that can generate a diverse set of interactive and controllable environments despite training from video-only data.

This model contains three key components: a latent action model, a video tokenizer, and a dynamics model.

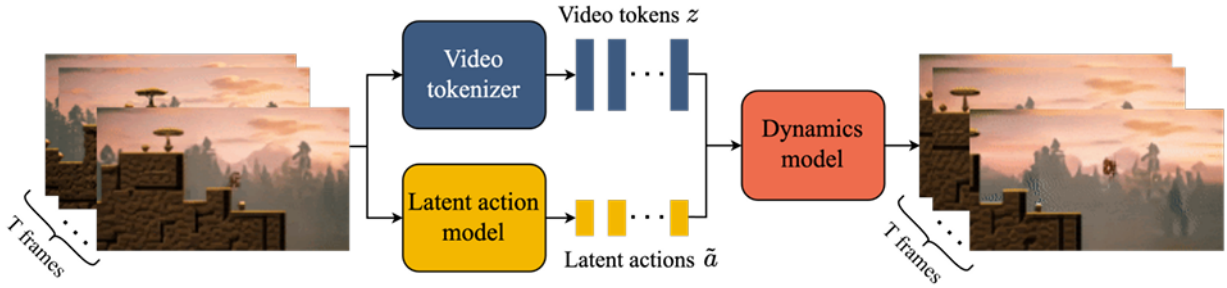

Figure 3 | **Genie model training**: Genie takes in T frames of video as input, tokenizes them into discrete tokens $\boldsymbol{z}$ via the video tokenizer, and infers the latent actions $\tilde{\boldsymbol{a}}$ between each frame with the latent action model. Both are then passed to the dynamics model to generate predictions for the next frames in an iterative manner.

Latent Action Model.

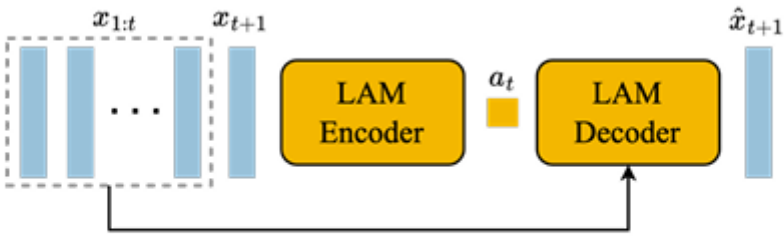

Figure 5 | **Latent action model**: learns actions a_t unsupervised from unlabelled video frames.

To train the model, they leverage a VQ-VAE based objective, which enables they to limit the number of predicted actions to a small discrete set of codes.

Video Tokenizer.

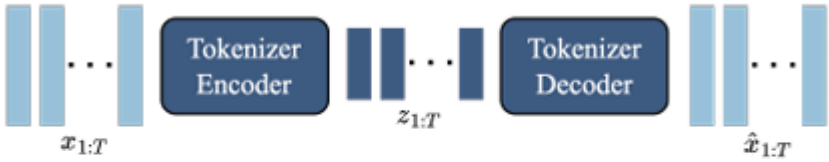

Figure 6 | **Video tokenizer**: a VQ-VAE with ST-transformer.

When they compress videos into discrete tokens they again make use of VQ-VAE. They also utilize the ST-transformer in both encoder and decoder to incoporate temporal dynamics in the encodings.

Dynamics Model

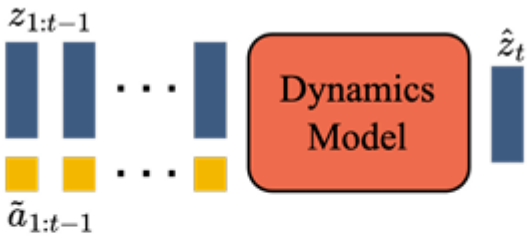

Figure 7 | **Dynamics model**: takes in video tokens and action embeddings, and predicts future masked video tokens.

The dynamics model is a decoder-only MaskGIT transformer. They again utilize an ST-transformer.

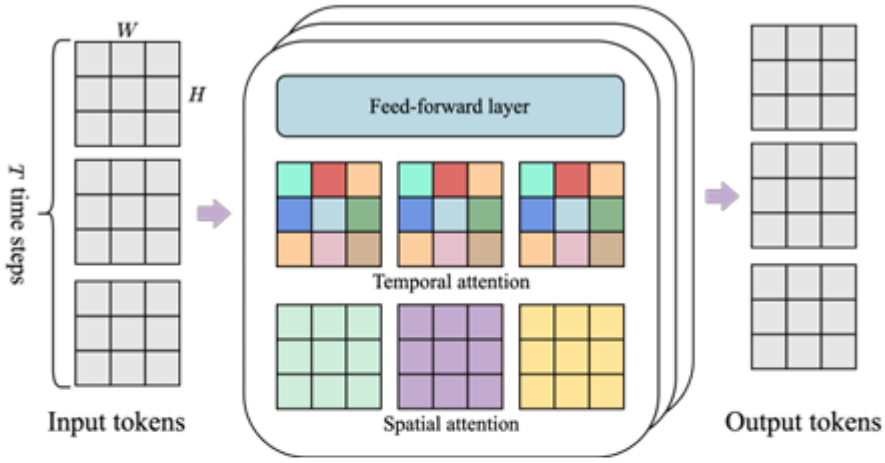

Figure 4 | **ST-transformer architecture**. The architecture is composed of L spatiotemporal blocks, each containing a spatial layer, temporal layer and feed-forward layer. Each color represents a single self-attention map, with the spatial layer attending over the $H \times W$ tokens from within a single time step, and temporal the same token from across the T time steps.